Satish Kumar

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Professional Summary

A Deep Learning enthusiast. Currently working on improving Deep Generative Learning algorithms like diffusion models and large language models(LLMs) by bringing state-of-the-art insights from Neuroscience . Energetic Software Engineer with 6+ years of experience having a diverse skill-set and creative drive to software application development. Proficient in writing code in various languages.And having extensive experience in developing customized Deep learning/Machine Learning architectures from scratch with good mathematical Intution & understanding.
My goal is simple "A complete understanding of how the brain works".

Employment History

Research Associate, Indian Institute of Technology Madras. Chennai, Tamil Nadu
Nov. 2023 - Apr. 2024

Projects

1.Real time video streaming based monitoring of breathing patterns in patients

Description:

Environment / Technology used: PyTorch,Python,OpenCV, , LangGraph, RAG,Knowledge Graph, Neo4J, DeepEval
Model: Lukas-kanade method of optical flow, Eulerian Video Magnification, LLMs, RAG, Multi-Modal RAG

Technical Lead- R&D, eNeuroLearn. Ann Arbor, Michigan
Mar. 2023 - Oct. 2023

Projects

Skills

- Langchain, LangGraph, Llama-Index, Neo4J
- Large Language Models(LLMs),Agentic AI, LLMs based Agents
- Generative AI, Knowlege Graph, RAG, Multi-Modal RAG
- DeepEval, Foolbox, Adversarial Robustness Toolbox (ART)
- Scikit Learn & Matlab
- PyTorch
- flask
- docker
- Stable Diffusion models, ControlNets, T2I-adapter, IP-adapter
- Tensorflow

Languages

- Hindi
- English

German

1.Developing GenAI applications using Satble Diffusion Model and Large Language Models(LLMs)

Description:

Environment / Technology used: PyTorch,Python,Langchain, LangGraph, Knowledge Graph,Neo4J, DeepEval
Model: Stable Diffusion, Large language models, RAG, Vector Database, Multi-Modal RAG,

IT Analyst, Neurotechnology Lab India pvt ltd. Gurgaon, Haryana
Dec. 2022 - Feb. 2023

Senior Software Engineer (R&D), Reverie Language Technologies Pvt Ltd. Bengaluru, Karnataka
August. 2019 - Jul. 2022

Projects
1.Gender Classifier from speech
Description:
Environment / Technology used: Keras,Python,DNNs
Feature type: fBank feature after applying VAD (Voice Activity Detection)
Model: 2-layer DNN with Dropout
Duration: Minimum duration of speech 0.75 seconds

2.Language Classifier from monologue for Indian Languages (API)
Description:
Environment / Technology used:Python,Keras
Feature type: I-vector feature of 500 dimension
Model: 2-layer DNN with Dropout

3.Voice activity detection system
Description:
Environment / Technology used:PyTorch,Python,DNNs
Feature type: fBank feature transformed as an images of 2-Dimensional to be fed to CNN model
Model: 3-layer CNN with Dropout

4.Speaker Diarization
Description:
Environment / Technology used:END-TO-END NEURAL SPEAKER DIARIZATION
Description:

- Prepared simulated data for training
- Multi-head self attention based model is used for training
- Analysed performance of model on diarization task on real dataset

5.Speech Separation

Description:

Environment / Technology used: ConvTasnet

- Prepared simulated data for training
- Model used in training is ConvTasnet
- Analysed performance of model on separation task on real dataset

6.ASR system for Marathi, Bengali & Punjabi languages

Description:

Environment / Technology used: Kaldi,Kenlm

Roles & Responsibilities:

- Developed,maintain and improve ASR system for Marathi, Bengali & Punjabi languages using open source framework Kaldi and Kenlm
- Domain specific data collection
- Pre-process speech data, using Kaldi toolkit
- Build test and trained data for ASR
- Training the ASR model
- Evaluate ASR output
- Provide linguistics knowledge on ASR output to improve accuracy

7.A Mixture of Expert system to Improve ASR

Description:

Environment / Technology used:PyTorch-Kaldi,Pytorch

Description:

- A 3-class phonetic classifier for Voiced,Unvoiced,Silence is introduced
- LSTM acoustic model is used

Data scientist, Real Time Signals Technologies Pvt. Ltd..
Bengaluru, Karnataka

Aug. 2018 - Jul. 2019

Project Associate, Learning and Extraction of Acoustic Patterns (LEAP) lab, Indian Institute of Science. Bengaluru, Karnataka

September. 2017 - July. 2018

Projects:

1.Participated in NIST LRE-17 challenge

Role Played:

Language Recognition model development based on i-vector features and Deep

Neural Network models.

Data collection for Indian languages.

Environment / Technology used: Tensorflow, Keras, Python, Scikit-

Learn, Matlab, Pandas
Description:

- Train a GMM-UBM based model to extract i-vectors for each speech utterances
- Trained a DNN network to classify i-vectors for each languages
- Analysed the performance of the system on different duration of speech utterances

2.Domain Adaptation of speech segments using Generative Adversarial Networks(GANs)

Role Played:
Developed a model which can adapt the the shorter duration of speech to a
longer duration ones.
Environment / Technology used: PyTorch, Python, Scikit-Learn, Matlab, Pandas,Adversarial Robustness Toolbox(ART), Foolbox
Description:

- Train a Generator-Discriminator based system for domain adaptation of i-vectors of different durations.

Presentations

- LabTalk on “Variational Treatment of Probabilistic Models” at LEAP Lab,IISc Bengaluru.

https://github.com/Satishpas2/Variational-Treatment-of-Probabilistic-Directed-Graphical-Models/blob/master/Var_Inference.pdf

- LabTalk on “Generative Adversarial Networks(GANs)” at LEAP Lab,IISc Bengaluru.

https://github.com/Satishpas2/Generative-Adversarial-Netwoeks-A-Tutorial/blob/master/GAN_PPT_LEAP.pdf

- Presentation on "Language Identification from Monologue" at Reverie Language Technologies,Bengaluru

Education

National Institute of Technology, Surathkal, Karnataka,India
Master of Technology, 2016, Communication Engineering, July. 2016

Uttar Pradesh Technical University, Lucknow, Uttar Pradesh,India
Batchelor of Technology, Electronics & Communication Engineering, June. 2008

References

Dr. K. P. Unnikrishnan

CO-FOUNDER & CHIEF SCIENTIST eNeuroLearn

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Google scholar profile: https://scholar.google.com/citations?user=4gng3xoAAAAJ&hl=en

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Personal Details

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Additional Information

Github Link:
https://github.com/Satishpas2
Linkedin Link:
https://www.linkedin.com/in/satish-kumar-571034109/

Publications

- B. Padi, S. Ramoji, V. Yeruva, S. Kumar and S. Ganapathy, "The LEAP Language Recognition System for LRE 2017 Challenge - Improvements and Error Analysis", Odyssey: The speaker and language recognition workshop, 2018.

https://www.isca-speech.org/archive/Odyssey_2018/pdfs/39.pdf